

Yujian Zhao

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EDUCATION

University of Maryland

Master of Science, Computer Science, GPA 3.754

University of Maryland

Bachelor of Science, Computer Science and Mathematics, GPA 3.693

- Dean's List: Fall 2020, Fall 2019, Fall 2018, Spring 2018, Fall 2017

College Park, MD, USA

Aug 2021 – May 2023

College Park, MD, USA

Aug 2017 – May 2021

TECHNICAL SKILLS

- Proficient Programming Language: Python, JavaScript, PostgreSQL, Java, C#
- Learned or used Language: C++, Ruby, OCaml, Dr. Racket, HTML, CSS, R, Coq
- Frameworks: Node JS, React JS, ASP.NET, Flask
- Database: MongoDB, MySQL, PostgreSQL
- Game Engine: Unity
- Visualization Tool: Tableau
- Packages: Pytorch, Tensorflow, Pandas, Scikit-Learn, Matplotlib
- Amazon Cloud Service: S3

WORK EXPERIENCE

Community Dreams Foundation

Python Developer

Maryland, US

July 2023 - Present

- Extracted over 80,000 data on financial institutions and utility companies from websites using BeautifulSoup and Selenium.
- Designed and managed PostgreSQL databases, boosting efficiency by 15% in data storage and retrieval.
- Created compelling Tableau dashboards to visualize real estate data across 10+ states, aiding informed decision-making.
- Forecasted real estate market data using advanced time series forecasting techniques with TensorFlow in Python, including LSTM and RNNs, achieving a 92% accuracy rate.
- Developed Python-based RESTful APIs with Flask, ensuring adherence to REST principles, robust API documentation, and seamless integration with front-end interfaces for an optimal user experience.

The University of Maryland, Department of Computer Science

Graduate Teaching Assistant

College Park, Maryland, US

Jan 2022 - Dec 2022

- Enhanced student learning through assisting 65 students in understanding parallel and multi-thread programming.
- Created test cases using Junit for assignment submission and held weekly office hours to help students debug.

Cylerian LLC

Software Engineer Intern

New Jersey, US

Jun 2022 - Aug 2022

- Built data pipelines to query data from the company's server and improved the data inquiry time by 5%.
- Tidied the over 15,000 raw log data into tables in pandas and made it easier for machine learning tasks.
- Developed LSTM models in Python to accurately detect anomalous user activities with an 83% success rate.
- Employed the TensorFlow JS models and created a Cron job to perform daily anomaly detection for the company.

City@UMD

Software Developer

College Park, Maryland, US

Aug 2020 - Jan 2022

- Built an API pipeline using JavaScript and SQL that allows uploading over 10,000 data every week.
- Architected and constructed a robust RESTful API in JavaScript, utilizing Node.js for server-side development and integrating it with a React.js front-end, ensuring efficient communication and a responsive user interface.
- Designed interfaces and visualized the data for 5 pages on the website using React JS.
- Constructed a database in MongoDB using Python for over 100,000 Wi-Fi data helps university to save energy.

PROJECT EXPERIENCE

The Effect of Count Sketch on a Neural Network's Robustness

Algorithms in Machine Learning: Guarantees and Analyses

Sept 2021 - Dec 2021

- Researched how count sketched neural networks performed compared to a standard neural network.
- Built different models using count-sketch, Topk, and t-Topk algorithms, and trained them using MNIST and Cifar datasets.
- Performed FSGM adversarial attack algorithm to test model robustness under PGD attack algorithm.
- Result: Count-sketch algorithm decreased the accuracy of the model trained by the MNIST dataset by 10 percent under adversarial attack with epsilon less than 0.7.

Social Biases in the Chinese Natural Language Dataset

Common-sense Reasoning and Natural Language Understanding

Aug 2022 - Dec 2022

- Used Pointwise Mutual Information methods to find out the potential gender and age biases in the Original Chinese Natural Language Inference dataset. Found potential biases for women and men.
- Trained GPT-2 using only premises of the Original Chinese Natural Language Inference dataset. Developed the re-trained model to generate sentences using prompts for each gender group. Biases and stereotypes were found in the dataset.

Evaluating the robustness of vulnerability exploit predictors to data distribution shifts

Master's Non-Thesis Paper

Aug 2022 - May 2023

- Evaluated vulnerability exploit predictors, specifically assessing the performance and feature importance of the Expected Exploitability (EE) metric. Compared, and analyzed two EE models trained during different periods.
- Identified key features contributing to accurate predictions and analyzed score changes over time.
- Provided insights for improving exploitability prediction models and supporting effective mitigation of software vulnerabilities.